Water storage change across Thailand

Climate reconstruction and the limits of human attribution

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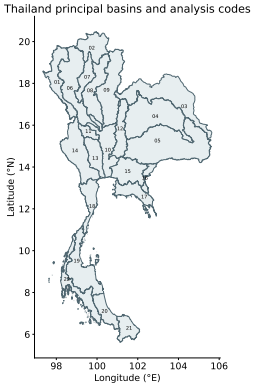
Research focus

- How are total and groundwater storage changing across Thailand’s 22 principal basins?
- How much of the observed change can the climate baseline explain?
- How strongly do independent observations support the remaining residual?

Research objectives

Objective	Output
1 Storage change	Monthly TWSA/GWSA, Sen slopes and Mann–Kendall tests
2 Climate and residual	Climate reconstruction, GLDAS benchmark and observation checks
3 Decision support	Interactive basin app, input data and reproducible code

Thailand’s 22 principal basins



- Official ONWR reporting framework
- GRACE information extends across basin boundaries
- All basin tables retain coverage and uncertainty

ONWR principal-basin geometry. Reporting units do not imply independent GRACE observations.

Data and scientific roles

Record	Role
CSR RL06.3 GRACE/GRACE-FO	Observed total storage
GLDAS Noah and G3P	Non-groundwater storage compartments
Paired WaterGAP simulations	Naturalized and historical-human baselines
Native P/PET and CHIRPS	Weather reconstruction and rainfall sensitivity
Thai wells and MSEA-Res	Groundwater and reservoir consistency evidence

Primary TWSA: April 2002–September 2024. Derived GWSA: April 2002–September 2023.

Analysis workflow

- Harmonize monthly units, dates, area weights and the 2004–2009 reference
- Calculate GWSA and estimate storage trends
- Reconstruct naturalized storage using weather and climate memory
- Calculate the residual, examine uncertainty and compare independent observations

Groundwater storage balance

$$\mathrm{GWSA} = \mathrm{TWSA} - \mathrm{SMSA} - \mathrm{SWSA} - \mathrm{CWSA} - \mathrm{SWEA}$$

- Subtract soil, surface, canopy and snow storage anomalies
- Use common valid cells and a common reference period
- Groundwater uncertainty includes every subtracted compartment

Storage anomalies use mm of equivalent water depth. Snow is small in Thailand but retained consistently.

Naturalized and historical simulations

Symbol	Meaning
O	Observed or compartment-derived storage
C	Naturalized WaterGAP baseline
A	Historical WaterGAP with represented human activities

Paired simulations share model structure and forcing. The naturalized baseline remains a model experiment.

Residual accounting

$$R = O - C = \underbrace{(A - C)}_{H} + \underbrace{(O - A)}_{E}$$

- H describes the human effect represented by the paired model
- E retains the observation–historical-model discrepancy
- The primary residual uses the published naturalized baseline

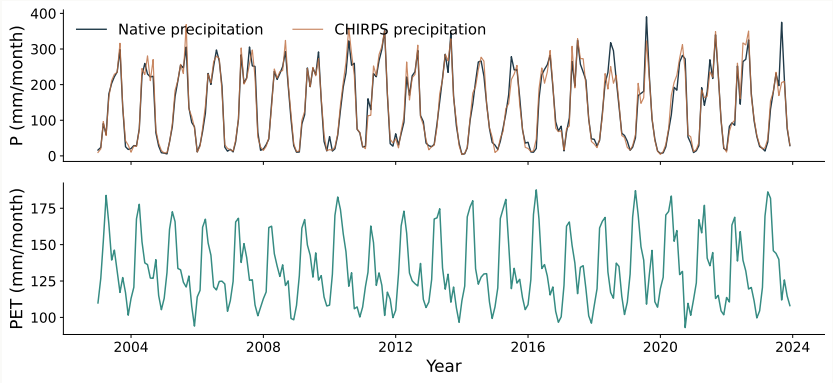
The statistical emulator reconstructs C from weather. Its prediction does not replace C in the primary residual.

Chronological model evaluation

Period	Years	Use
Warm-up	1981–1990	Initialize climate memory
Training	1991–2009	Fit transformations and tune models
Validation	2010–2015	Freeze model selection
Test	2016–2023	Evaluate unseen years

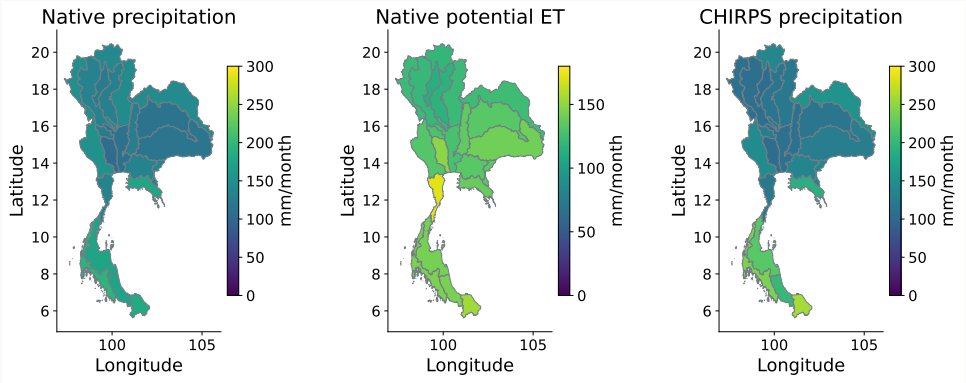
Ridge, Elastic Net, splines and Random Forest. Predictors exclude GRACE and groundwater observations.

Precipitation and potential ET



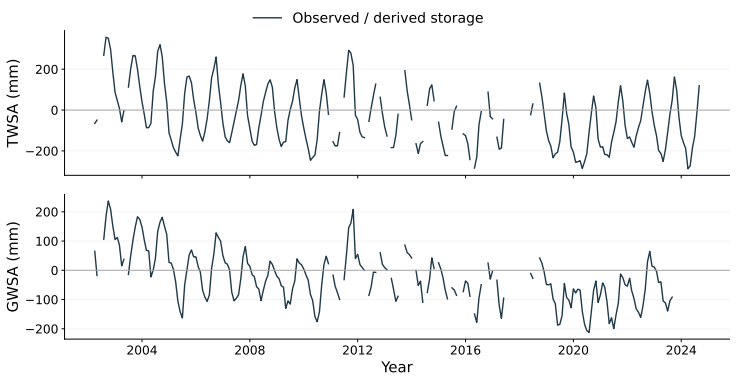
National monthly values, 2003–2023. Native P/PET drive reconstruction. CHIRPS supplies the rainfall replacement test.

Spatial distribution of the meteorological inputs



Available-month basin means, 2003–2023. Values are mm/month, not annual totals. Each map uses its labelled scale.

Observed storage through time



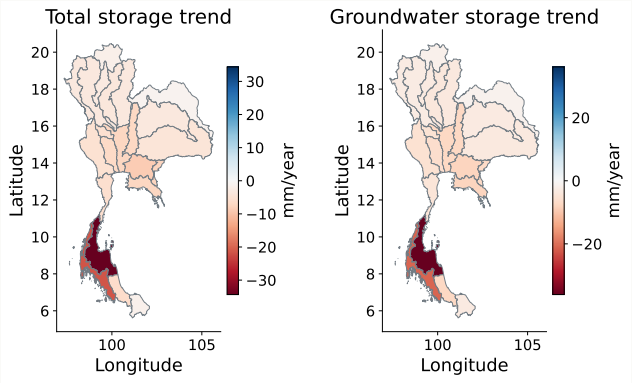
National CSR TWSA and compartment-derived GWSA. Missing months remain gaps. Groundwater ends earlier because of compartment availability.

National storage decline

Storage	Sen slope (mm/year)	Conditional 95% interval
Total storage	−7.51	−9.79 to −5.68
Groundwater storage	−6.84	−8.77 to −5.11

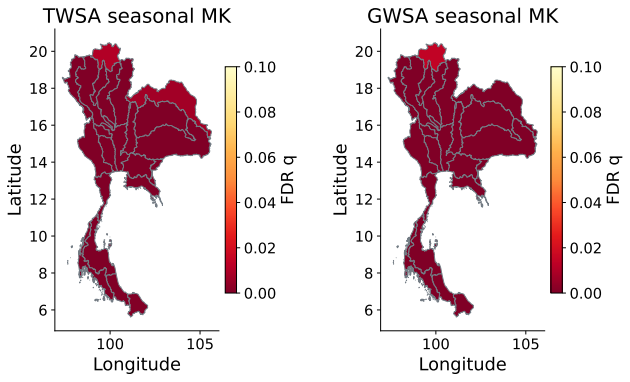
237 TWSA months through September 2024; 225 GWSA months through September 2023. Full-record seasonal Sen slopes.

Storage trend patterns across basins



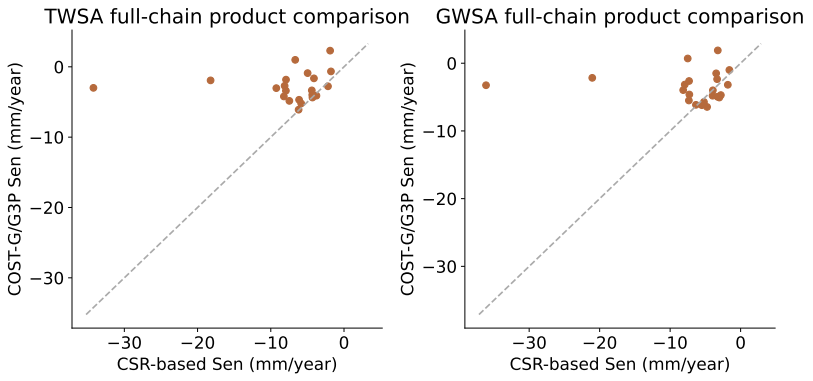
Seasonal Sen slopes. Negative values indicate decline. All basins retain the CSR small-area caution.

Statistical evidence for basin trends



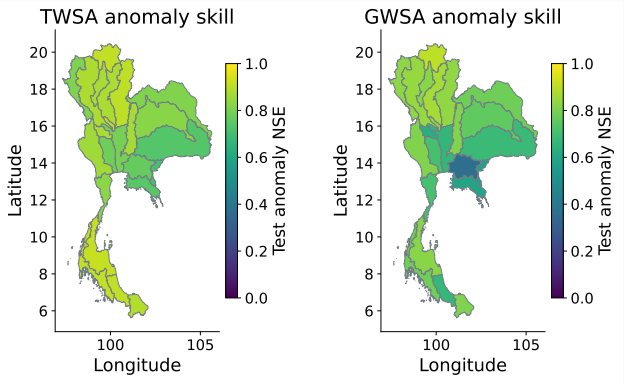
Seasonal Mann–Kendall with basin FDR. Colours saturate at q = 0.10. This test does not remove serial dependence or product uncertainty.

Basin ranking depends on the processing chain



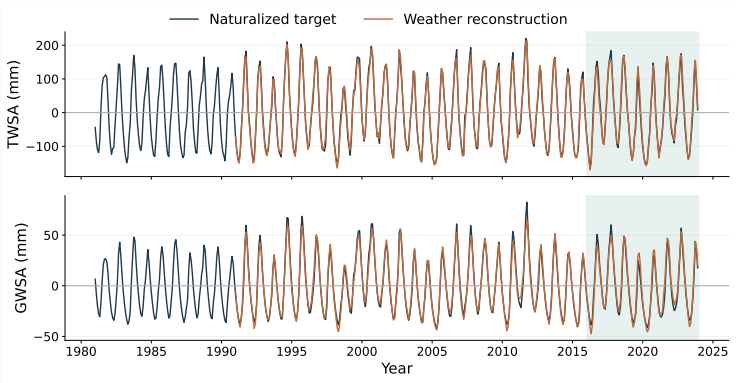
Common-date rank correlations: TWSA 0.138 and GWSA 0.027. Groundwater chains also differ in compartment processing.

Climate reconstruction performance



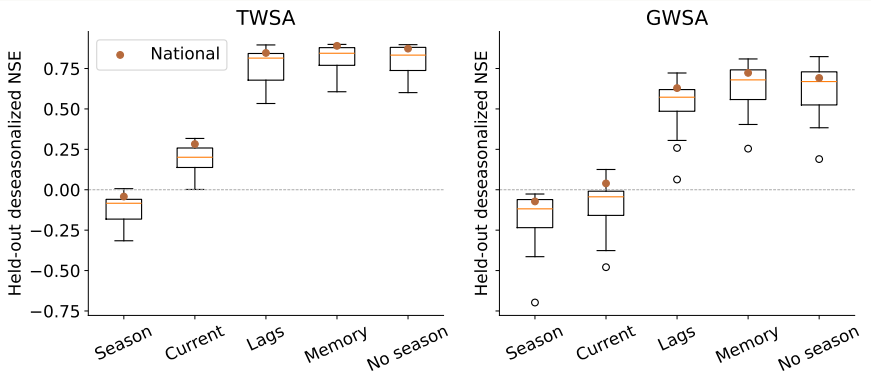
Ensemble test anomaly NSE against naturalized WaterGAP, 2016–2023. High scores verify reconstruction of the parent model.

Naturalized storage and weather reconstruction



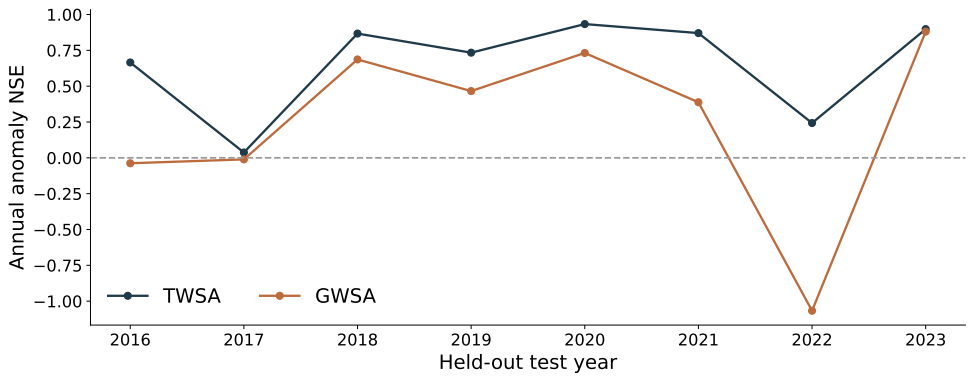
National series. Training: 1991–2009; validation: 2010–2015; test: 2016–2023. Parent-model agreement is distinct from observational validation.

Climate memory improves reconstruction



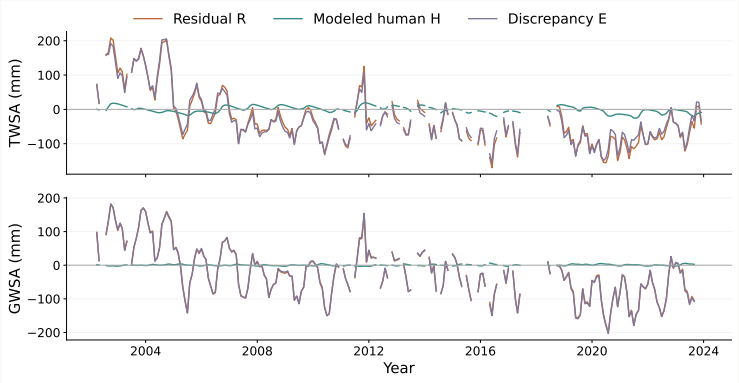
Controlled Ridge experiment. National test anomaly NSE: current weather / memory = 0.283 / 0.890 for TWSA and 0.039 / 0.723 for GWSA.

Performance varies between test years



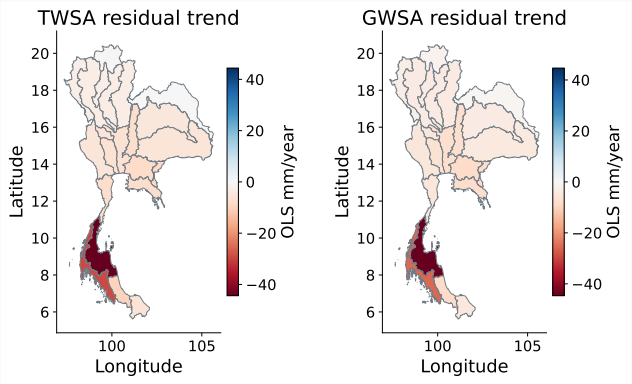
2022 national anomaly NSE: TWSA 0.234; GWSA −1.067. Negative NSE indicates performance below the mean benchmark.

Residual and modeled human effects differ



R = H + E on common observed months. The remaining discrepancy includes unresolved model and observation errors.

Spatial residual trends



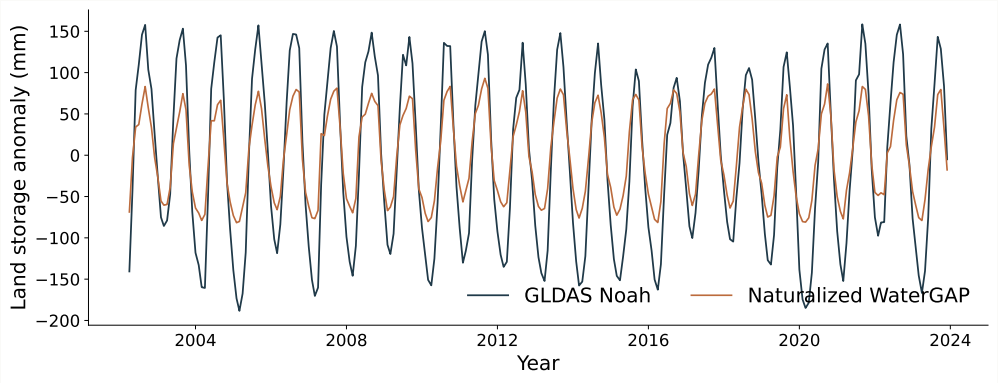
Calendar-month fixed-effect OLS slopes on matched support. These slopes describe residual change and do not measure pumping.

Additive national trend partition

Component	TWSA	GWSA
Observed O	-8.36	-7.28
Naturalized C	-0.54	-0.21
Modeled human H	-0.49	0.06
Residual R	-7.81	-7.07
Discrepancy E	-7.32	-7.13

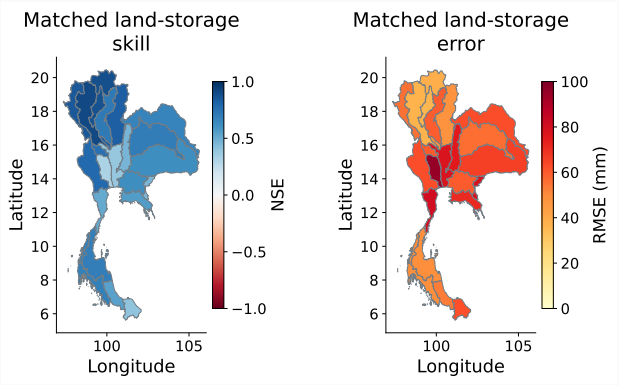
OLS slopes, mm/year, using the same months. R = H + E. Seasonal Sen slopes are reported separately because their medians are not additive.

GLDAS comparison on matching compartments



Soil + canopy + snow only. National r = 0.969, NSE = 0.738, RMSE = 52.49 mm across 261 months. Groundwater and surface water are excluded.

GLDAS agreement varies among basins



Matched land-storage benchmark. NSE colours saturate below −1 and RMSE above 100 mm. GLDAS also contributes to the groundwater subtraction.

Groundwater level and storage use different units

Observation	What the comparison establishes
Groundwater head or depth (m)	Local aquifer response and direction
Regional storage anomaly (mm)	Water-mass change over a basin
Paired annual correlation	Agreement in timing on observed years

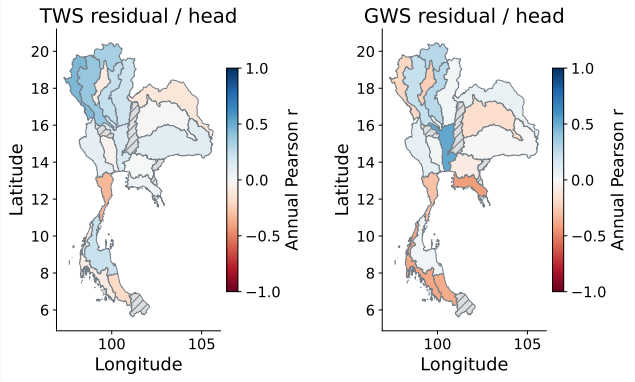
A head-to-storage conversion requires aquifer storage coefficients and representative areas. Those inputs are unavailable here.

Well selection changes residual agreement

Station panel	TWS r	GWS r	GWS years
Longest overlap	0.127	0.037	17
Fixed eligibility / wider history	0.657	0.511	17
Common period	0.325	0.029	8
Strictly balanced	0.305	≈0.000	8

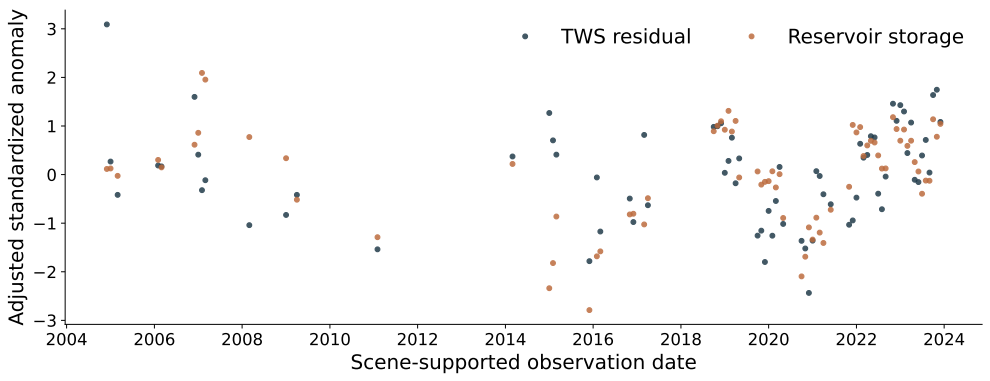
511 wells provide 5,484 aggregates. Co-located wells are grouped. The common-period panels cover eight usable storage years within 2013–2022.

Spatial groundwater consistency



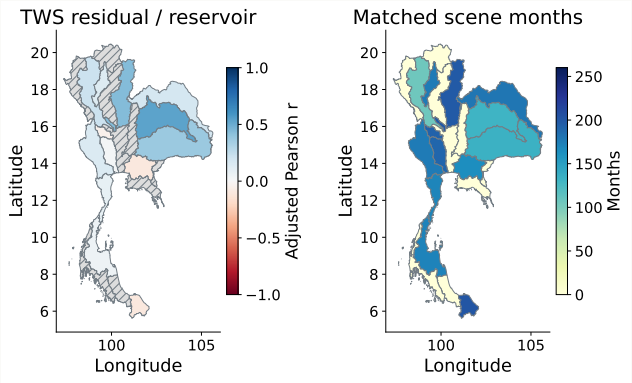
Residual–head correlations on available annual pairs. Grey hatching marks insufficient evidence. These comparisons cannot validate a human-only total.

Independent reservoir evidence



26-reservoir fixed panel, 73 scene-supported months. Adjusted r = 0.439, year-cluster 95% interval 0.188–0.665. Both curves remove season and linear trend.

Reservoir agreement and observation support



Scene-screened MSEA-Res Level 1. Reservoir storage checks a surface-water component. Climate and operating decisions both affect reservoirs.

Sensitivity to the rainfall source

Test anomaly NSE in the frozen Ridge memory experiment

Storage	Native rainfall	CHIRPS replacement
TWSA	0.890	0.688
GWSA	0.723	0.562

PET and the native-forced parent target stay fixed. This tests forcing dependence, rather than which rainfall product is more accurate.

Decision support and reproducibility

- Interactive storage histories, trend maps and observation support for all 22 basins
- Downloadable basin performance tables and spatial/temporal forcing figures
- Input subsets, pinned dependencies and executable scientific workflows

https://pratikkhanal16.github.io/thailand-water-storage-dss/extended/ • Source repository: pratikkhanal16/thailand-water-storage-dss

Conclusions

- The primary records show national TWSA and GWSA decline
- Basin rankings and residual interpretation depend on products and observation support
- Climate memory reconstructs the naturalized parent well, with weaker performance in some years
- Wells and reservoirs provide complementary, limited evidence for residual interpretation

Contribution: a reproducible joint assessment of product sensitivity, climate memory, paired-model discrepancy and monitoring-network uncertainty.

Selected references

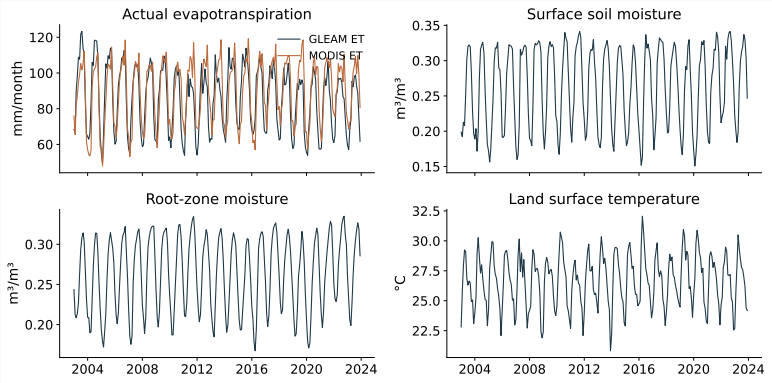
- Save et al. (2016), CSR mascons. doi:10.1002/2016JB013007
- Müller Schmied et al. (2024), WaterGAP. doi:10.5194/gmd-17-8817-2024
- Sharifi et al. (2025), groundwater storage. doi:10.5194/hess-29-6985-2025
- MSEA-Res (2025), reservoir storage. doi:10.5194/essd-17-2693-2025
- GROW groundwater observations. doi:10.5281/zenodo.15149480

Full bibliography, source terms and basin results accompany the report. Theme: Metropolis by Matthias Vogelgesang and contributors, https://github.com/matze/mtheme.

Supporting results

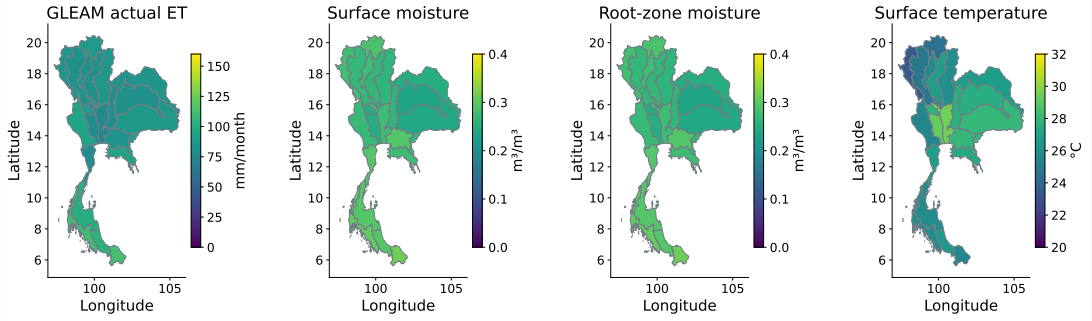
- Additional hydroclimatic context
- Basin performance tables for questions
- Observation support and interpretation

Supporting hydroclimatic histories



National monthly observations, 2003–2023. Actual ET, soil moisture and surface temperature provide context; they are not added primary predictors.

Spatial hydroclimatic context



Basin available-month means, 2003–2023. Product coverage differs. Volumetric moisture is distinct from storage in mm.

Spatial support and uncertainty

- National aggregation touches 68 native CSR mascons
- Inverse weight concentration is about 50.1 and does not count independent observations
- Every principal basin is below the provider’s approximate 200,000 km² small-area caution
- Conditional intervals do not include all gravity, compartment and hydrological-model errors

Observed storage trends for every basin (01–11)

Basin	TWS slope	TWS q	n	GWS slope	GWS q	n
01	-3.59	<.001	237	-2.88	<.001	225
02	-1.90	0.010	237	-1.83	0.014	225
03	-1.32	0.007	237	-1.59	<.001	225
04	-3.65	<.001	237	-3.32	<.001	225
05	-4.42	<.001	237	-3.47	<.001	225
06	-3.71	<.001	237	-3.10	<.001	225
07	-3.21	<.001	237	-3.32	<.001	225
08	-3.79	<.001	237	-3.97	<.001	225
09	-3.74	<.001	237	-3.94	<.001	225
10	-7.36	<.001	237	-7.47	<.001	225
11	-5.50	<.001	237	-5.20	<.001	225

Slopes are seasonal Sen estimates in mm/year. q is seasonal Mann–Kendall FDR across basins; n is observed months. Full intervals and original p-values remain in CSR_basin_trends.csv. Basin names are listed in Appendix A.

Observed storage trends for every basin (12–22)

Basin	TWS slope	TWS q	n	GWS slope	GWS q	n
12	-7.06	<.001	237	-7.35	<.001	225
13	-6.89	<.001	237	-6.36	<.001	225
14	-5.25	<.001	237	-5.48	<.001	225
15	-8.64	<.001	237	-8.12	<.001	225
16	-7.68	<.001	237	-7.30	<.001	225
17	-7.51	<.001	237	-7.92	<.001	225
18	-5.85	<.001	237	-4.75	<.001	225
19	-34.37	<.001	237	-36.08	<.001	225
20	-6.25	<.001	237	-7.45	<.001	225
21	-1.62	<.001	237	-3.16	<.001	225
22	-21.48	<.001	237	-20.95	<.001	225

Slopes are seasonal Sen estimates in mm/year. q is seasonal Mann–Kendall FDR across basins; n is observed months. Full intervals and original p-values remain in CSR_basin_trends.csv. Basin names are listed in Appendix A.

TWSA climate reconstruction performance (01–11)

Basin	n	r	NSE	Anom NSE	RMSE	MAE
01	96	0.991	0.981	0.807	22.08	16.90
02	96	0.990	0.979	0.904	17.32	13.88
03	96	0.990	0.973	0.812	27.37	19.39
04	96	0.981	0.961	0.821	15.61	11.58
05	96	0.974	0.946	0.732	21.43	15.40
06	96	0.991	0.981	0.876	15.90	12.00
07	96	0.987	0.974	0.868	15.24	11.08
08	96	0.987	0.972	0.873	15.38	11.43
09	96	0.992	0.983	0.901	14.14	10.83
10	96	0.967	0.928	0.792	42.23	30.13
11	96	0.990	0.980	0.875	15.34	12.27

Frozen ensemble, 2016–2023 test data. Target: naturalized WaterGAP, not GRACE. Errors are mm; anomaly NSE removes the training monthly climatology. Higher NSE and lower errors are better.

TWSA climate reconstruction performance (12–22)

Basin	n	r	NSE	Anom NSE	RMSE	MAE
12	96	0.987	0.974	0.864	11.22	8.35
13	96	0.974	0.942	0.778	32.00	23.99
14	96	0.991	0.981	0.856	18.49	14.13
15	96	0.975	0.942	0.764	28.65	22.72
16	96	0.979	0.957	0.705	14.60	11.45
17	96	0.987	0.972	0.742	16.75	13.62
18	96	0.979	0.957	0.827	14.38	10.97
19	96	0.982	0.965	0.911	17.38	13.55
20	96	0.982	0.963	0.903	22.54	18.01
21	96	0.977	0.951	0.887	25.44	18.76
22	96	0.985	0.970	0.903	17.36	13.16

Frozen ensemble, 2016–2023 test data. Target: naturalized WaterGAP, not GRACE. Errors are mm; anomaly NSE removes the training monthly climatology. Higher NSE and lower errors are better.

GWSA climate reconstruction performance (01–11)

Basin	n	r	NSE	Anom NSE	RMSE	MAE
01	96	0.982	0.963	0.830	5.74	4.47
02	96	0.986	0.956	0.901	4.98	4.08
03	96	0.982	0.953	0.781	10.59	7.70
04	96	0.969	0.933	0.755	7.04	5.33
05	96	0.963	0.927	0.678	8.87	6.32
06	96	0.980	0.958	0.842	3.94	2.92
07	96	0.977	0.948	0.824	4.83	3.96
08	96	0.984	0.963	0.865	5.52	4.22
09	96	0.982	0.958	0.832	6.04	4.78
10	96	0.968	0.935	0.661	4.90	4.09
11	96	0.958	0.914	0.627	7.09	5.40

Frozen ensemble, 2016–2023 test data. Target: naturalized WaterGAP, not GRACE. Errors are mm; anomaly NSE removes the training monthly climatology. Higher NSE and lower errors are better.

GWSA climate reconstruction performance (12–22)

Basin	n	r	NSE	Anom NSE	RMSE	MAE
12	96	0.981	0.962	0.812	4.62	3.61
13	96	0.959	0.920	0.703	5.50	4.45
14	96	0.985	0.970	0.812	4.39	3.12
15	96	0.963	0.902	0.360	10.51	8.46
16	96	0.970	0.933	0.564	10.88	8.87
17	96	0.980	0.952	0.589	11.67	9.60
18	96	0.958	0.907	0.711	6.84	5.11
19	96	0.968	0.926	0.823	10.40	8.46
20	96	0.946	0.892	0.668	11.69	9.70
21	96	0.953	0.904	0.808	9.19	7.33
22	96	0.967	0.922	0.780	11.32	8.94

Frozen ensemble, 2016–2023 test data. Target: naturalized WaterGAP, not GRACE. Errors are mm; anomaly NSE removes the training monthly climatology. Higher NSE and lower errors are better.

GLDAS matched land storage performance (01–11)

Basin	n	r	NSE	RMSE	MAE	Bias
01	261	0.949	0.850	55.42	46.06	4.24
02	261	0.955	0.877	38.82	31.16	-3.20
03	261	0.957	0.670	62.72	55.91	3.44
04	261	0.940	0.702	56.19	48.08	11.12
05	261	0.931	0.620	65.72	57.34	12.09
06	261	0.961	0.901	37.15	30.56	3.75
07	261	0.962	0.874	36.09	29.37	-2.62
08	261	0.926	0.712	59.38	51.34	-2.20
09	261	0.956	0.825	48.01	41.12	1.04
10	261	0.871	0.390	77.84	69.70	7.74
11	261	0.900	0.756	59.88	49.64	3.91

Soil, canopy and snow only; naturalized WaterGAP compared with GLDAS Noah on matched months. Errors are mm; bias = WaterGAP minus GLDAS. This does not validate full TWS, GWS or human-only storage.

GLDAS matched land storage performance (12–22)

Basin	n	r	NSE	RMSE	MAE	Bias
12	261	0.848	0.465	75.38	66.62	7.09
13	261	0.814	0.317	95.06	85.91	13.97
14	261	0.955	0.788	62.51	53.84	4.43
15	261	0.934	0.621	60.08	51.92	5.74
16	261	0.860	0.453	78.69	68.66	9.02
17	261	0.891	0.559	71.30	62.24	3.90
18	261	0.858	0.514	80.68	71.40	13.50
19	261	0.866	0.690	49.28	39.89	0.01
20	261	0.844	0.544	53.29	41.99	10.73
21	261	0.678	0.403	62.21	51.62	9.14
22	261	0.887	0.686	49.60	40.29	-6.04

Soil, canopy and snow only; naturalized WaterGAP compared with GLDAS Noah on matched months. Errors are mm; bias = WaterGAP minus GLDAS. This does not validate full TWS, GWS or human-only storage.

Residual agreement with annual groundwater head (01–11)

Basin	TWS r	TWS detr r	Years	GWS r	GWS detr r	Years
01	0.449	0.582	9	-0.215	0.167	8
02	0.330	0.345	17	0.279	0.399	16
03	-0.121	-0.084	9	0.063	-0.109	8
04	-0.009	-0.030	9	-0.168	-0.023	8
05	0.113	0.103	9	-0.007	0.070	8
06	0.385	0.432	15	0.229	0.293	14
07	-0.089	-0.043	9	-0.244	0.047	8
08	0.236	0.157	9	0.277	-0.124	8
09	0.208	0.176	9	0.026	-0.198	8
10	0.152	0.050	9	0.519	0.200	8
11	—	—	5	—	—	4

Longest-overlap retained-flags panel. Positive r means lower residual tends to occur with lower head. Detr r removes a linear trend. An em dash means unavailable evidence, not zero. Head is not converted into storage; no cross-unit NSE or RMSE is reported. Other station panels and conditional intervals remain in the complete CSV.

Residual agreement with annual groundwater head (12–22)

Basin	TWS r	TWS detr r	Years	GWS r	GWS detr r	Years
12	—	—	5	—	—	4
13	-0.044	-0.212	17	0.027	-0.147	16
14	0.094	0.074	17	0.093	0.116	16
15	0.001	-0.041	9	-0.082	-0.187	8
16	—	—	3	—	—	2
17	0.041	0.136	9	-0.420	-0.215	8
18	-0.335	-0.394	9	-0.289	-0.115	8
19	0.227	0.096	9	0.033	-0.147	8
20	-0.197	-0.656	18	-0.375	-0.722	17
21	—	—	0	—	—	0
22	-0.080	-0.143	15	-0.389	-0.450	14

Longest-overlap retained-flags panel. Positive r means lower residual tends to occur with lower head. Detr r removes a linear trend. An em dash means unavailable evidence, not zero. Head is not converted into storage; no cross-unit NSE or RMSE is reported. Other station panels and conditional intervals remain in the complete CSV.

Complete basin results and source data

- Report Appendix C: 11 tables for every basin
- Report Appendix D: 10 spatial and temporal forcing figures
- The website supplies full-precision CSVs, vector figures and runnable code

https://pratikkhanal16.github.io/thailand-water-storage-dss/extended/basin_forcing/